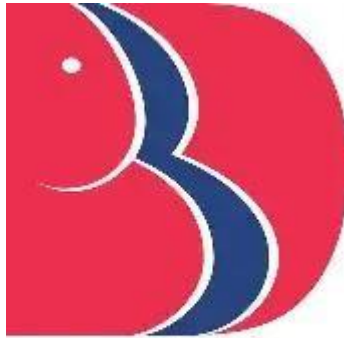


Babu Banarasi Das University



NO SQL PROJECT ON

Student Record Management System

Submitted to: Mr. Harendra Singh

Submitted by: Aman

Department: BCA(DS & AI)

Roll_no: 1250258068

OBJECTIVES:

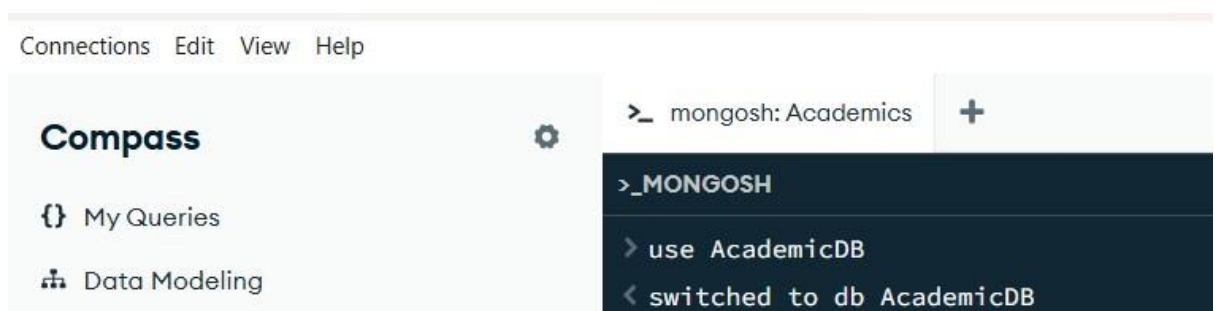
- To understand the concept of **NoSQL databases**.
- To create a **database and collection** in MongoDB.
- To insert and manage **50 student records**.
- To perform **CRUD operations** — Create, Read, Update, and Delete.
- To use **comparison operators** such as \$gt, \$gte, \$lt, \$lte, \$ne, and \$in.
- To perform **sorting and filtering** of student records.

- To perform **search and retrieval operations** based on specific conditions.
- To understand and implement **AND (\$and)** and **OR (\$or)** operations.
- To understand how MongoDB can be used to **efficiently manage and retrieve student information**.

1. CREATE DATABASE

MongoDB Query:

○ *use AcademicDB*



2.CREATE COLLECTION

MongoDB Query:

○ `db.createCollection("students")`

```
> db.createCollection("students")
< { ok: 1 }
```

2. INSERT 50 STUDENTS

MongoDB Query:

○ `db.students.insertMany([ ])` `db.students.insertMany([`

```
{roll:1, name:           ", age:19,marks:88,city:"Lucknow"},
{roll:2, name:"Abhishek Dixit", age:18,marks:76,city:"Jaunpur"}, {roll:3,
name:"Shivangi Tiwari", age:19,marks:92,city:"Kanpur"}, {roll:4,
name:"Ankush Yadav", age:18,marks:69,city:"Ayodhya"},
{roll:5, name:"Vanshika Dixit", age:19,marks:81,city:"Varanasi"},
{roll:6, name:"Abhishk Maurya", age:20,marks:85,city:"Lucknow"},
{roll:7, name: "Anchal Tiwari", age:19,marks:73,city:"Prayagraj"},
{roll:8, name: "Pratik Dubey", age:19,marks:90,city:"Lucknow"},
{roll:9, name: "Siya Aggarwal", age:20,marks:78,city:"Gorakhpur"},
{roll:10, name: "Vivek Maurya", age:18,marks:65,city:"Kanpur"},
{roll:11, name: "Rajveer Singh", age:20,marks:71,city:"Jaunpur"},
{roll:12, name: "Shreya Tripathi", age:19,marks:95,city:"Lucknow"},
```

{roll:13, name: "Akshita Gupta", age:18,marks:84,city:"Ayodhya"},
{roll:14, name:" Shruti Pandey", age:19,marks:87,city:"Prayagraj"},
{roll:15, name: "Adarsh Singh", age:20,marks:74,city:"Varanasi"},
{roll:16, name: "Sejal Mishra", age:18,marks:89,city:"Lucknow"},
{roll:17, name: "Astha Pandey", age:19,marks:82,city:"Kanpur"},
{roll:18, name: "Jaya Tiwari", age:18,marks:58,city:"Gonda"},
{roll:19, name: "Harshita", age:20,marks:91,city:"Lucknow"},
{roll:20, name: "Omkar", age:19,marks:63,city:"Bareilly"},
{roll:21, name: "Ayush", age:18,marks:77,city:"Lucknow"},
{roll:22, name: "Saumya", age:19,marks:86,city:"Jaunpur"},
{roll:23, name: "Vidya", age:20,marks:72,city:"Kanpur"},
{roll:24, name: "Adarsh", age:19,marks:80,city:"Ayodhya"},
{roll:25, name: "Anmol", age:18,marks:67,city:"Prayagraj"},
{roll:26, name: "Anushka", age:20,marks:94,city:"Lucknow"},
{roll:27, name: "Ananya", age:19,marks:88,city:"Varanasi"},
{roll:28, name: "Bhaskar", age:18,marks:61,city:"Gorakhpur"},
{roll:29, name: "Rohan", age:19,marks:75,city:"Kanpur"},
{roll:30, name: "Nishant", age:20,marks:79,city:"Lucknow"},
{roll:31, name: "Vineet", age:18,marks:68,city:"Jaunpur"}, {roll:32,
name: "Anika", age:19,marks:90,city:"Lucknow"},
{roll:33, name: "Jai", age:20,marks:55,city:"Ayodhya"},
{roll:34, name: "Lakshya", age:18,marks:83,city:"Kanpur"},
{roll:35, name: "Krishna", age:19,marks:96,city:"Prayagraj"},
{roll:36, name: "Rumana", age:20,marks:70,city:"Lucknow"},
{roll:37, name: "Priyanshu", age:18,marks:81,city:"Varanasi"},
{roll:38, name: "Aditya", age:19,marks:66,city:"Gorakhpur"},
{roll:39, name: "Dhruv", age:20,marks:93,city:"Lucknow"},
{roll:40, name: "Nivedita", age:19,marks:85,city:"Kanpur"},

```
{roll:41, name: "Meenu", age:18,marks:74,city:"Jaunpur"},  
{roll:42, name: "Akanksha", age:20,marks:89,city:"Lucknow"},  
{roll:43, name: "Sarika",age:19,marks:62,city:"Ayodhya"},  
{roll:44, name: "Anchal Tiwari",age:18,marks:78,city:"Prayagraj"},  
{roll:45, name: "Umesh Bajpai", age:20,marks:84,city:"Lucknow"},  
{roll:46, name: "Priya Shukla", age:19,marks:73,city:"Kanpur"},  
{roll:47, name: "Aman", age:18,marks:59,city:"Gonda"},  
{roll:48, name: "Priya", age:20,marks:87,city:"Lucknow"},  
{roll:49, name: "Rahul", age:19,marks:64,city:"Bareilly"},  
{roll:50, name: "Neha", age:18,marks:91,city:"Varanasi"}  
)
```

CONNECTIONS (1)

Search connections

- Adarsh
 - admin
 - config
 - local

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

- Adarsh
 - admin
 - config
 - local

```
> db.students.insertMany([
  {roll:1, name:"Adarsh Dixit", age:19,marks:88,city:"Lucknow"},
  {roll:2, name:"Abhishek Dixit", age:18,marks:76,city:"Jaunpur"},
  {roll:3, name:"Shivangi Tiwari", age:19,marks:92,city:"Kanpur"},
  {roll:4, name:"Ankush Yadav", age:18,marks:69,city:"Ayodhya"},
  {roll:5, name:"Vanshika Dixit", age:19,marks:81,city:"Varanasi"},
  {roll:6, name:"Abhishek Maurya", age:20,marks:85,city:"Lucknow"},
  {roll:7, name:"Anchal Tiwari", age:19,marks:73,city:"Prayagraj"},
  {roll:8, name:"Pratik Dubey", age:19,marks:90,city:"Lucknow"},
  {roll:9, name:"Siya Aggarwal", age:20,marks:78,city:"Gorakhpur"},
  {roll:10, name:"Vivek Maurya", age:18,marks:65,city:"Kanpur"},
  {roll:11, name:"Rajveer Singh", age:20,marks:71,city:"Jaunpur"},
  {roll:12, name:"Shreya Tripathi", age:19,marks:95,city:"Lucknow"},
  {roll:13, name:"Akshita Gupta", age:18,marks:84,city:"Ayodhya"},
  {roll:14, name:"Shruti Pandey", age:19,marks:87,city:"Prayagraj"},
  {roll:15, name:"Adarsh Singh", age:20,marks:74,city:"Varanasi"},
  {roll:16, name:"Sejal Mishra", age:18,marks:89,city:"Lucknow"},
  {roll:17, name:"Astha Pandey", age:19,marks:82,city:"Kanpur"},
  {roll:18, name:"Jaya Tiwari", age:18,marks:58,city:"Gonda"},
  {roll:19, name:"Harshita", age:20,marks:91,city:"Lucknow"},
  {roll:20, name:"Omkar", age:19,marks:63,city:"Bareilly"},
  {roll:21, name:"Ayush", age:18,marks:77,city:"Lucknow"},
  {roll:22, name:"Saumya", age:19,marks:86,city:"Jaunpur"},
  {roll:23, name:"Vidya", age:20,marks:72,city:"Kanpur"},
  {roll:24, name:"Adarsh", age:19,marks:80,city:"Ayodhya"},
  {roll:25, name:"Anmol", age:18,marks:67,city:"Prayagraj"},
  {roll:26, name:"Anushka", age:20,marks:94,city:"Lucknow"},
  {roll:27, name:"Ananya", age:19,marks:88,city:"Varanasi"},
  {roll:28, name:"Bhaskar", age:18,marks:61,city:"Gorakhpur"},
  {roll:29, name:"Rohan", age:19,marks:75,city:"Kanpur"},
  {roll:30, name:"Nishant", age:20,marks:79,city:"Lucknow"},
  {roll:31, name:"Vineet", age:18,marks:68,city:"Jaunpur"},
  {roll:32, name:"Anika", age:19,marks:90,city:"Lucknow"},
  {roll:33, name:"Jai", age:20,marks:55,city:"Ayodhya"},
  {roll:34, name:"Lakshya", age:18,marks:83,city:"Kanpur"},
  {roll:35, name:"Krishna", age:19,marks:96,city:"Prayagraj"},
  {roll:36, name:"Rumana", age:20,marks:70,city:"Lucknow"},
  {roll:37, name:"Priyanshu", age:18,marks:81,city:"Varanasi"},
  {roll:38, name:"Aditya", age:19,marks:66,city:"Gorakhpur"},
  {roll:39, name:"Dhruv", age:20,marks:93,city:"Lucknow"},
  {roll:40, name:"Nivedita", age:19,marks:85,city:"Kanpur"},
  {roll:41, name:"Meenu", age:18,marks:74,city:"Jaunpur"},
  {roll:42, name:"Akanksha", age:20,marks:89,city:"Lucknow"},
  {roll:43, name:"Sarika", age:19,marks:62,city:"Ayodhya"},
  {roll:44, name:"Anchal Tiwari", age:18,marks:78,city:"Prayagraj"},
  {roll:45, name:"Umesh Bajpai", age:20,marks:84,city:"Lucknow"},
  {roll:46, name:"Priya Shukla", age:19,marks:73,city:"Kanpur"},
  {roll:47, name:"Aman", age:18,marks:59,city:"Gonda"},
  {roll:48, name:"Priya", age:20,marks:87,city:"Lucknow"},
  {roll:49, name:"Rahul", age:19,marks:64,city:"Bareilly"},
  {roll:50, name:"Neha", age:18,marks:91,city:"Varanasi"}
])
```

```
{roll:23, name:"Vidya", age:20,marks:72,city:"Kanpur"},
{roll:24, name:"Adarsh", age:19,marks:80,city:"Ayodhya"},
{roll:25, name:"Anmol", age:18,marks:67,city:"Prayagraj"},
{roll:26, name:"Anushka", age:20,marks:94,city:"Lucknow"},
{roll:27, name:"Ananya", age:19,marks:88,city:"Varanasi"},
{roll:28, name:"Bhaskar", age:18,marks:61,city:"Gorakhpur"},
{roll:29, name:"Rohan", age:19,marks:75,city:"Kanpur"},
{roll:30, name:"Nishant", age:20,marks:79,city:"Lucknow"},
{roll:31, name:"Vineet", age:18,marks:68,city:"Jaunpur"},
{roll:32, name:"Anika", age:19,marks:90,city:"Lucknow"},
{roll:33, name:"Jai", age:20,marks:55,city:"Ayodhya"},
{roll:34, name:"Lakshya", age:18,marks:83,city:"Kanpur"},
{roll:35, name:"Krishna", age:19,marks:96,city:"Prayagraj"},
{roll:36, name:"Rumana", age:20,marks:70,city:"Lucknow"},
{roll:37, name:"Priyanshu", age:18,marks:81,city:"Varanasi"},
{roll:38, name:"Aditya", age:19,marks:66,city:"Gorakhpur"},
{roll:39, name:"Dhruv", age:20,marks:93,city:"Lucknow"},
{roll:40, name:"Nivedita", age:19,marks:85,city:"Kanpur"},
{roll:41, name:"Meenu", age:18,marks:74,city:"Jaunpur"},
{roll:42, name:"Akanksha", age:20,marks:89,city:"Lucknow"},
{roll:43, name:"Sarika", age:19,marks:62,city:"Ayodhya"},
{roll:44, name:"Anchal Tiwari", age:18,marks:78,city:"Prayagraj"},
{roll:45, name:"Umesh Bajpai", age:20,marks:84,city:"Lucknow"},
{roll:46, name:"Priya Shukla", age:19,marks:73,city:"Kanpur"},
{roll:47, name:"Aman", age:18,marks:59,city:"Gonda"},
{roll:48, name:"Priya", age:20,marks:87,city:"Lucknow"},
{roll:49, name:"Rahul", age:19,marks:64,city:"Bareilly"},
{roll:50, name:"Neha", age:18,marks:91,city:"Varanasi"}
])
```

4. DISPLAY ALL STUDENTS

MongoDB Query:

○ `db.students.find()`

```
> db.students.find()
< {
  _id: ObjectId('6ab3fb5d3768a611a133f8b7'),
  roll: 1,
  name: 'Adarsh Dixit',
  age: 19,
  marks: 88,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ab3fb5d3768a611a133f8b8'),
  roll: 2,
  name: 'Abhishek Dixit',
  age: 18,
  marks: 76,
  city: 'Jaunpur'
}
{
  _id: ObjectId('6ab3fb5d3768a611a133f8b9'),
  roll: 3,
  name: 'Shivangi Tiwari',
  age: 19,
  marks: 92,
  city: 'Kanpur'
}
```

5. EQUAL TO — \$eq MongoDB

Query:

○ `db.students.find({marks: {$eq: 75}})`

```
> db.students.find({marks: {$eq: 75}})
< {
  _id: ObjectId('6ab3fb5d3768a611a133f8d3'),
  roll: 29,
  name: 'Rohan',
  age: 19,
  marks: 75,
  city: 'Kanpur'
}
```

6. GREATER THAN — \$gt

MongoDB Query:

○ `db.students.find({marks: {$gt: 60}})`

Explanation: Finds students whose marks are greater than 60.

```
> db.students.find({marks: {$gt: 60}})
< {
  _id: ObjectId('6ab3fb5d3768a611a133f8b7'),
  roll: 1,
  name: 'Adarsh Dixit',
  age: 19,
  marks: 88,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ab3fb5d3768a611a133f8b8'),
  roll: 2,
  name: 'Abhishek Dixit',
  age: 18,
  marks: 76,
  city: 'Jaunpur'
}
```

7. LESS THAN — \$lt MongoDB

Query:

○ `db.students.find({marks: {$lt: 60}})`

```
> db.students.find({marks: {$lt: 60}})
< {
  _id: ObjectId('6ab3fb5d3768a611a133f8c8'),
  roll: 18,
  name: 'Jaya Tiwari',
  age: 18,
  marks: 58,
  city: 'Gonda'
}
{
  _id: ObjectId('6ab3fb5d3768a611a133f8d7'),
  roll: 33,
  name: 'Jai',
  age: 20,
  marks: 55,
  city: 'Ayodhya'
}
{
  _id: ObjectId('6ab3fb5d3768a611a133f8e5'),
  roll: 47,
  name: 'Aman',
  age: 18,
  marks: 59,
  city: 'Gonda'
}
Students >
```

8. GREATER THAN OR EQUAL

TO — \$gte MongoDB Query:

○ *db.students.find({marks: {\$gte: 90}})*

```
> db.students.find({marks: {$gte: 90}})
< {
  _id: ObjectId('6ab3fb5d3768a611a133f8b9'),
  roll: 3,
  name: 'Shivangi Tiwari',
  age: 19,
  marks: 92,
  city: 'Kanpur'
}
{
  _id: ObjectId('6ab3fb5d3768a611a133f8be'),
  roll: 8,
  name: 'Pratik Dubey',
  age: 19,
  marks: 90,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ab3fb5d3768a611a133f8c2'),
  roll: 12,
  name: 'Shreya Tripathi',
  age: 19,
  marks: 95,
  city: 'Lucknow'
}
```

9. LESS THAN OR EQUAL TO

— **\$lte** MongoDB Query:

db.students.find({marks: {\$lte: 70}})

```
> db.students.find({marks: {$lte: 70}})
< {
  _id: ObjectId('6ab3fb5d3768a611a133f8ba'),
  roll: 4,
  name: 'Ankush Yadav',
  age: 18,
  marks: 69,
  city: 'Ayodhya'
}
{
  _id: ObjectId('6ab3fb5d3768a611a133f8c0'),
  roll: 10,
  name: 'Vivek Maurya',
  age: 18,
  marks: 65,
  city: 'Kanpur'
}
{
  _id: ObjectId('6ab3fb5d3768a611a133f8c8'),
  roll: 18,
  name: 'Jaya Tiwari',
  age: 18,
  marks: 58,
  city: 'Gonda'
}
```

10. UPDATE OPERATION

MongoDB Query:

```
○ db.students.updateOne(  
  {roll:26},  
  {$set:{marks:89}}  
)
```

```
> db.students.updateOne({roll: 26}, {$set: {marks: 89}})  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}  
Students >
```

```
> db.students.find({roll: 26})  
< {  
  _id: ObjectId('6ab3fb5d3768a611a133f8d0'),  
  roll: 26,  
  name: 'Anushka',  
  age: 20,  
  marks: 89,  
  city: 'Lucknow'  
}  
Students >
```

11. DELETE ONE OPERATION

MongoDB Query:

```
○ db.students.deleteOne({roll:10})
```

```
> db.students.deleteOne({roll: 10})  
< {  
  acknowledged: true,  
  deletedCount: 1  
}  
Students > |
```

12. DELETE MANY OPERATION

MongoDB Query:

```
db.students.deleteMany(  
{ marks: {$lt:60}})
```

```
> db.students.find({ $and: [ {marks: {$gt: 80}}, {age: 19} ] })  
< {  
  _id: ObjectId('6ab3fb5d3768a611a133f8b7'),  
  roll: 1,  
  name: 'Adarsh Dixit',  
  age: 19,  
  marks: 88,  
  city: 'Lucknow'  
}  
{  
  _id: ObjectId('6ab3fb5d3768a611a133f8b9'),  
  roll: 3,  
  name: 'Shivangi Tiwari',  
  age: 19,  
  marks: 92,  
  city: 'Kanpur'  
}  
{  
  _id: ObjectId('6ab3fb5d3768a611a133f8bb'),  
  roll: 5,  
  name: 'Vanshika Dixit',  
  age: 19,  
  marks: 81,  
  city: 'Varanasi'  
}
```

13. AND OPERATION — \$and

MongoDB Query:

```
○ db.students.find({  
  $and:[  
    {marks:{$gt:80}},  
    {age:21}  
  ]  
})
```

```
> db.students.find({ $and: [ {marks: {$gt: 80}}, {age: 21} ] })  
<  
Students >
```

14. OR OPERATION — \$or

MongoDB Query:

```
○ db.students.find({  
  $or:[  
    {city:"Lucknow"},  
    {city:"Gorakhpur"} ]})
```

```
> db.students.find({ $or: [ {city: "Lucknow"}, {city: "Gorakhpur"} ] })  
< [  
  {  
    _id: ObjectId('6ab3fb5d3768a611a133f8b7'),  
    roll: 1,  
    name: 'Adarsh Dixit',  
    age: 19,  
    marks: 88,  
    city: 'Lucknow'  
  },  
  {  
    _id: ObjectId('6ab3fb5d3768a611a133f8bc'),  
    roll: 6,  
    name: 'Abhishk Maurya',  
    age: 20,  
    marks: 85,  
    city: 'Lucknow'  
  },  
  {  
    _id: ObjectId('6ab3fb5d3768a611a133f8be'),  
    roll: 8,  
    name: 'Pratik Dubey',  
    age: 19,  
    marks: 90,  
    city: 'Lucknow'  
  }  
]
```

15. SORT OPERATION

MongoDB Ascending order Query:

○ `db.students.find().sort({marks: 1})`

```
> db.students.find().sort({marks: 1})
< {
  _id: ObjectId('6ab3fb5d3768a611a133f8d2'),
  roll: 28,
  name: 'Bhaskar',
  age: 18,
  marks: 61,
  city: 'Gorakhpur'
}
{
  _id: ObjectId('6ab3fb5d3768a611a133f8e1'),
  roll: 43,
  name: 'Sarika',
  age: 19,
  marks: 62,
  city: 'Ayodhya'
}
{
  _id: ObjectId('6ab3fb5d3768a611a133f8ca'),
  roll: 20,
  name: 'Omkar',
  age: 19,
  marks: 63,
  city: 'Bareilly'
}
```

B. MongoDB Descending Order Query:

○ `db.students.find().Sort ({marks: -1})`

```
> db.students.find().sort({marks: -1})
< {
  _id: ObjectId('6ab3fb5d3768a611a133f8d9'),
  roll: 35,
  name: 'Krishna',
  age: 19,
  marks: 96,
  city: 'Prayagraj'
}
{
  _id: ObjectId('6ab3fb5d3768a611a133f8c2'),
  roll: 12,
  name: 'Shreya Tripathi',
  age: 19,
  marks: 95,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ab3fb5d3768a611a133f8dd'),
  roll: 39,
  name: 'Dhruv',
  age: 20,
  marks: 93,
  city: 'Lucknow'
}
```

16. LIMIT OPERATION MongoDB

Query:

- `db.students.find().limit(5)`

```
> db.students.find().limit(5)
< {
  _id: ObjectId('6ab3fb5d3768a611a133f8b7'),
  roll: 1,
  name: 'Adarsh Dixit',
  age: 19,
  marks: 88,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ab3fb5d3768a611a133f8b8'),
  roll: 2,
  name: 'Abhishek Dixit',
  age: 18,
  marks: 76,
  city: 'Jaunpur'
}
{
  _id: ObjectId('6ab3fb5d3768a611a133f8b9'),
  roll: 3,
  name: 'Shivangi Tiwari',
  age: 19,
  marks: 92,
  city: 'Kanpur'
}
```

17. SEARCH BY CITY MongoDB

Query:

- `db.students.find({city:"Lucknow"})`

```
> db.students.find({city:"Lucknow"})
< {
  _id: ObjectId('6ab3fb5d3768a611a133f8b7'),
  roll: 1,
  name: 'Adarsh Dixit',
  age: 19,
  marks: 88,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ab3fb5d3768a611a133f8bc'),
  roll: 6,
  name: 'Abhishk Maurya',
  age: 20,
  marks: 85,
  city: 'Lucknow'
}
{
  _id: ObjectId('6ab3fb5d3768a611a133f8be'),
  roll: 8,
  name: 'Pratik Dubey',
  age: 19,
  marks: 90,
  city: 'Lucknow'
}
```

18. COUNT DOCUMENTS

MongoDB Query:

- `db.students.countDocuments()`

```
> db.students.countDocuments()
< 46
```

19.UPDATE MANY OPERATION

MongoDB Query:

- `db.students.updateMany({city: "Lucknow"}, {$set: {state: "UP"}})`

```
> db.students.updateMany({city: "Lucknow"}, {$set: {state: "UP"}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 15,
  modifiedCount: 15,
  upsertedCount: 0
}
```

```
> db.students.findOne({roll: 5})
< {
  _id: ObjectId('6ab3fb5d3768a611a133f8bb'),
  roll: 5,
  name: 'Vanshika Dixit',
  age: 19,
  marks: 81,
  city: 'Varanasi'
}
```

20.DISTINCT

MongoDB Query:

- `db.students.distinct("city")`

```
> db.students.distinct("city")
< [
  'Ayodhya', 'Bareilly',
  'Gorakhpur', 'Jaunpur',
  'Kanpur', 'Lucknow',
  'Prayagraj', 'Varanasi'
]
```